

Litly - Movie Review Application

A full-stack mobile application built with React Native (Expo) frontend and NestJS backend for discovering and reviewing movies.

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Project Overview

Litly is a cross-platform movie review application that allows users to:

- Browse and discover movies
- Create and manage user accounts with authentication
- Write and read reviews for movies
- Access an admin dashboard for content management

The application consists of two main parts:

- **Frontend:** React Native mobile app using Expo
- **Backend:** NestJS REST API with PostgreSQL database

Tech Stack

Frontend

- **React Native** 0.81.5 - Mobile app framework
- **Expo** 54.0.0 - React Native development platform
- **React Navigation** 7.x - Navigation management
- **TailwindCSS** 3.4.19 - Utility-first CSS framework
- **NativeWind** 4.2.3 - Tailwind CSS for React Native
- **Zod** 4.3.6 - Schema validation
- **React Hook Form** 7.72.0 - Form state management
- **React Query** 5.95.2 - Data fetching and caching
- **Axios** 1.13.6 - HTTP client
- **Zustand** 5.0.12 - State management
- **i18next** 25.10.9 - Internationalization
- **TypeScript** ~5.9.2 - Type safety

Backend

- **NestJS** 11.0.1 - Progressive Node.js framework
- **TypeORM** 0.3.27 - Object-relational mapping
- **PostgreSQL** - Database
- **Bcrypt** 6.0.0 - Password hashing
- **Swagger** 11.2.0 - API documentation
- **Jest** 30.0.0 - Testing framework
- **ESLint & Prettier** - Code quality and formatting

Project Structure

```

LitlyReactNat/
├── src/                                # Frontend source code
│   ├── api/                          # API integration
│   │   ├── apiClient.ts              # Axios configuration
│   │   └── movie.hooks.ts            # React Query hooks
│   ├── components/                   # Reusable UI components
│   │   ├── MovieCard.tsx
│   │   └── ReviewItem.tsx
│   ├── navigation/                  # Navigation setup
│   │   ├── AppNavigator.tsx
│   │   └── types.ts
│   ├── schemas/                     # Validation schemas
│   │   ├── auth.schema.ts
│   │   └── movie.schema.ts
│   ├── screens/                     # Screen components
│   │   ├── AddReviewScreen.tsx
│   │   ├── AdminDashboardScreen.tsx
│   │   ├── HomeScreen.tsx
│   │   ├── LoginScreen.tsx
│   │   ├── MovieDetailsScreen.tsx
│   │   └── SignupScreen.tsx
│   └── store/                       # State management
│       └── useAuthStore.ts
├── backend/                          # Backend source code
│   ├── src/
│   │   ├── app.module.ts            # Root module
│   │   ├── main.ts                  # Application entry point
│   │   ├── database/                # Database configuration
│   │   ├── movies/                  # Movies module
│   │   ├── reviews/                # Reviews module
│   │   └── users/                   # Users module (auth)
│   ├── test/                        # E2E tests
│   └── package.json
├── assets/                           # Images and icons
├── App.tsx                           # Root component
├── app.json                           # Expo configuration
├── package.json                       # Frontend dependencies
├── tsconfig.json
├── tailwind.config.js
├── nativewind-env.d.ts
├── babel.config.js
└── metro.config.js

```

Backend Module Structure

- **app.module.ts** - Root module that imports all feature modules
- **database/** - Database setup and seeding
 - database.module.ts - Database configuration
 - seed.service.ts - Initial data seeding
- **users/** - User authentication and management
 - users.service.ts - User business logic
 - users.controller.ts - API endpoints
 - dto/ - Data transfer objects
 - entities/ - Database entities
- **movies/** - Movie management
 - movies.service.ts - Movie business logic
 - movies.controller.ts - API endpoints
 - entities/ - Movie entity
- **reviews/** - Review management
 - reviews.service.ts - Review business logic
 - reviews.controller.ts - API endpoints
 - entities/ - Review entity

Prerequisites

- **Node.js** >= 18.0.0
- **npm** or **pnpm** (pnpm recommended)
- **Expo CLI** (for mobile development)
- **PostgreSQL** (for backend database)
- **Android Studio** or **Xcode** (for mobile emulation)

Installation & Setup

1. Clone the Repository

```
git clone <repository-url>
cd LitlyReactNat
```

2. Install Frontend Dependencies

```
npm install
# or
pnpm install
```

3. Install Backend Dependencies

```
cd backend
npm install
# or
pnpm install
cd ..
```

4. Environment Configuration

Create a `.env` file in the backend directory with the following variables:

```
# Database Configuration
DB_HOST=localhost
DB_PORT=5432
DB_USERNAME=postgres
DB_PASSWORD=your_password
DB_NAME=litly

# JWT Configuration (optional)
JWT_SECRET=your_secret_key

# API Configuration
API_PORT=3000
API_URL=http://localhost:3000
```

5. Database Setup

```
cd backend

# Create PostgreSQL database
createdb litly

# Run migrations (if available)
npm run typeorm:migration:run

# Seed initial data (optional)
npm run seed
```

Running the Project

Option 1: Development Mode (Recommended)

Terminal 1 - Backend:

```
cd backend
npm run start:dev
```

The backend will start on `http://localhost:3000`

Terminal 2 - Frontend:

```
npm start
```

Then choose your platform:

- Press `i` for iOS
- Press `a` for Android
- Press `w` for Web

Option 2: Production Build

Backend:

```
cd backend
npm run build
npm run start:prod
```

Frontend:

```
npm run android
# or
npm run ios
# or
npm run web
```

Available Scripts

Frontend Scripts

```
# Start development server
npm run start

# Run on Android emulator
npm run android

# Run on iOS simulator
npm run ios

# Run on web browser
npm run web
```

Backend Scripts

```
# Start in development mode with watch
npm run start:dev

# Start in debug mode
npm run start:debug

# Build for production
npm run build

# Start production build
npm run start:prod

# Run linting
npm run lint

# Format code
npm run format

# Run unit tests
npm run test

# Run tests in watch mode
npm run test:watch

# Run tests with coverage
npm run test:cov

# Run E2E tests
npm run test:e2e
```

🔖 Features

User Authentication

- User registration and login
- Secure password hashing with bcrypt
- Token-based authentication
- Persistent authentication state with Zustand

Movie Management

- Browse available movies
- View movie details and ratings
- Search and filter movies
- Admin dashboard for movie management

Reviews System

- Write and submit reviews for movies
- Read reviews from other users
- Rate movies
- View review history

Admin Features

- Content management dashboard
- User management
- Movie catalog management
- Review moderation

Internationalization

- Multi-language support using i18next
- Language persistence in local storage

UI/UX

- Cross-platform compatibility (iOS, Android, Web)
- Responsive design with TailwindCSS

- Form validation with Zod and React Hook Form
- Smooth animations with React Native Reanimated

API Documentation

The backend API documentation is available via Swagger UI when the server is running:

```
http://localhost:3000/api/docs
```

Main API Endpoints

Users:

- POST /users/register - Register new user
- POST /users/login - User login
- GET /users/:id - Get user details
- PUT /users/:id - Update user profile

Movies:

- GET /movies - Get all movies
- GET /movies/:id - Get movie details
- POST /movies - Create movie (admin)
- PUT /movies/:id - Update movie (admin)
- DELETE /movies/:id - Delete movie (admin)

Reviews:

- GET /reviews - Get all reviews
- GET /reviews/movie/:movieId - Get reviews for a movie
- POST /reviews - Create review
- PUT /reviews/:id - Update review
- DELETE /reviews/:id - Delete review

Testing

Run Backend Tests

```
# Unit tests
cd backend
npm run test

# Watch mode
npm run test:watch

# Coverage report
npm run test:cov

# E2E tests
npm run test:e2e
```

Troubleshooting

Frontend Issues

Expo CLI not found:

```
npm install -g expo-cli
```

Port already in use:

```
# Change port in your system or kill the process
# On Windows:
netstat -ano | findstr :8081
taskkill /PID <PID> /F
```

Dependencies issue:

```
rm -rf node_modules package-lock.json
npm install
```

Backend Issues

PostgreSQL connection error:

- Ensure PostgreSQL is running
- Verify database credentials in .env
- Check database exists: `psql -l`

Port 3000 already in use:

```
# Kill process on port 3000
# On Windows:
netstat -ano | findstr :3000
taskkill /PID <PID> /F
```

Module not found errors:

```
cd backend
npm install
npm run build
```

Code Quality

Formatting

```
cd backend
npm run format
```

Linting

```
cd backend
npm run lint
```

Contributing

1. Fork the repository
2. Create a feature branch (`git checkout -b feature/amazing-feature`)
3. Commit changes (`git commit -m 'Add amazing feature'`)
4. Push to branch (`git push origin feature/amazing-feature`)
5. Open a Pull Request

License

This project is licensed under the UNLICENSED license.

Authors

- Your Name - Initial work

Support

For support, email support@litly.com or create an issue in the repository.

Roadmap

- ☐ Dark mode support
- ☐ Push notifications
- ☐ Social sharing features
- ☐ Advanced search filters
- ☐ User profile customization
- ☐ Movie recommendations engine
- ☐ Offline mode support
- ☐ Performance optimizations